

PRACTICAL WEEK1

TASK1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user. Accept a Linux command as input. Create a child process using `fork()`. Execute the command in the child process using an appropriate `exec()` system call. Allow the parent process to wait for the child using `wait()`. Display the Process ID (PID) of both the parent and child processes.

```
jhansi_priya@Jhansi-Priya: ~/ × + ▾
jhansi_priya@Jhansi-Priya:~$ mkdir skill1
jhansi_priya@Jhansi-Priya:~$ cd skill1
jhansi_priya@Jhansi-Priya:~/skill1$ nano command.c
jhansi_priya@Jhansi-Priya:~/skill1$ gcc command.c -o command
jhansi_priya@Jhansi-Priya:~/skill1$ ./command
Enter a Linux command: ls

Child Process
Child PID: 409
command command.c

Parent Process
Parent PID: 408
jhansi_priya@Jhansi-Priya:~/skill1$ |
```

```
GNU nano 7.2 command.c *
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0)
    {
        printf("Fork failed.\n");
        return 1;
    }

    else if (pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());

        execlp(command, command, NULL);

        printf("Command execution failed.\n");
    }

    else
    {
        wait(NULL);

        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
    }

    return 0;
}
```

```
jhansi_priya@Jhansi-Priya:~/skill11$ uname
Linux
jhansi_priya@Jhansi-Priya:~/skill11$ uname -a
Linux Jhansi-Priya 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
jhansi_priya@Jhansi-Priya:~/skill11$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                39 bits physical, 48 bits virtual
Byte Order:                   Little Endian
CPU(s):                       8
On-line CPU(s) list:         0-7
Vendor ID:                    GenuineIntel
Model name:                   11th Gen Intel(R) Core(TM) i7-1185G
                             7 @ 3.00GHz
CPU family:                   6
Model:                        140
Thread(s) per core:          2
Core(s) per socket:           4
Socket(s):                    1
```

```
jhansi_priya@Jhansi-Priya: ~/skill1$ lsblk
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	0	356.9M	1	disk	

```
jhansi_priya@Jhansi-Priya: ~/ x + v

Srbds: P
Tsa: Not affected
Tsx async abort: Not affected
Vmsscape: Not affected
jhansi_priya@Jhansi-Priya:~/skill1$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 356.9M 1 disk
sdb 8:16 0 159.4M 1 disk
sdc 8:32 0 2G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk /mnt/wslg/distro

jhansi_priya@Jhansi-Priya:~/skill1$ ps
PID TTY TIME CMD
289 pts/0 00:00:00 bash
444 pts/0 00:00:00 ps

jhansi_priya@Jhansi-Priya:~/skill1$ top
top - 13:42:32 up 5 min, 1 user, load average: 0.00, 0.00, 0.
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 z
%Cpu(s): 0.0 us, 0.1 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi,
MiB Mem : 7635.6 total, 5958.6 free, 504.1 used, 1323.
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7131.

PID USER PR NI VIRT RES SHR S %CPU %MEM
1 root 20 0 21876 13304 9824 S 0.0 0.2
2 root 20 0 3180 2204 2072 S 0.0 0.0
7 root 20 0 3220 2152 2008 S 0.0 0.0
53 root 19 -1 66840 19620 18504 S 0.0 0.3
99 root 20 0 25152 6568 5196 S 0.0 0.1
155 systemd+ 20 0 21344 13216 11092 S 0.0 0.2
159 systemd+ 20 0 91036 8040 7076 S 0.0 0.1
165 root 20 0 4244 2748 2496 S 0.0 0.0
```

```
jhansi_priya@Jhansi-Priya: ~/ x + v

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155 systemd+ 20 0 21344 13216 11092 S 0.0 0.2
159 systemd+ 20 0 91036 8040 7076 S 0.0 0.1
165 root 20 0 4244 2748 2496 S 0.0 0.0
166 message+ 20 0 9548 5412 4712 S 0.0 0.1
173 root 20 0 17984 8904 7860 S 0.0 0.1
175 root 20 0 1756104 13180 11016 S 0.0 0.2
187 syslog 20 0 222516 6004 4624 S 0.0 0.1
200 root 20 0 3124 2036 1900 S 0.0 0.0
206 root 20 0 107032 23220 13764 S 0.0 0.3
287 root 20 0 3184 1108 980 S 0.0 0.0
288 root 20 0 3200 1248 1108 S 0.0 0.0
289 jhansi_+ 20 0 6080 5396 3672 S 0.0 0.1
290 root 20 0 6696 4724 3944 S 0.0 0.1
333 jhansi_+ 20 0 20324 11572 9448 S 0.0 0.1
334 jhansi_+ 20 0 21168 3576 1848 S 0.0 0.0
356 jhansi_+ 20 0 6080 5340 3672 S 0.0 0.1
445 jhansi_+ 20 0 9284 5728 3560 R 0.0 0.1
```

